ESP Siesta Clock with Matrix Display

Technical Reference - English

2026 by HexOhm

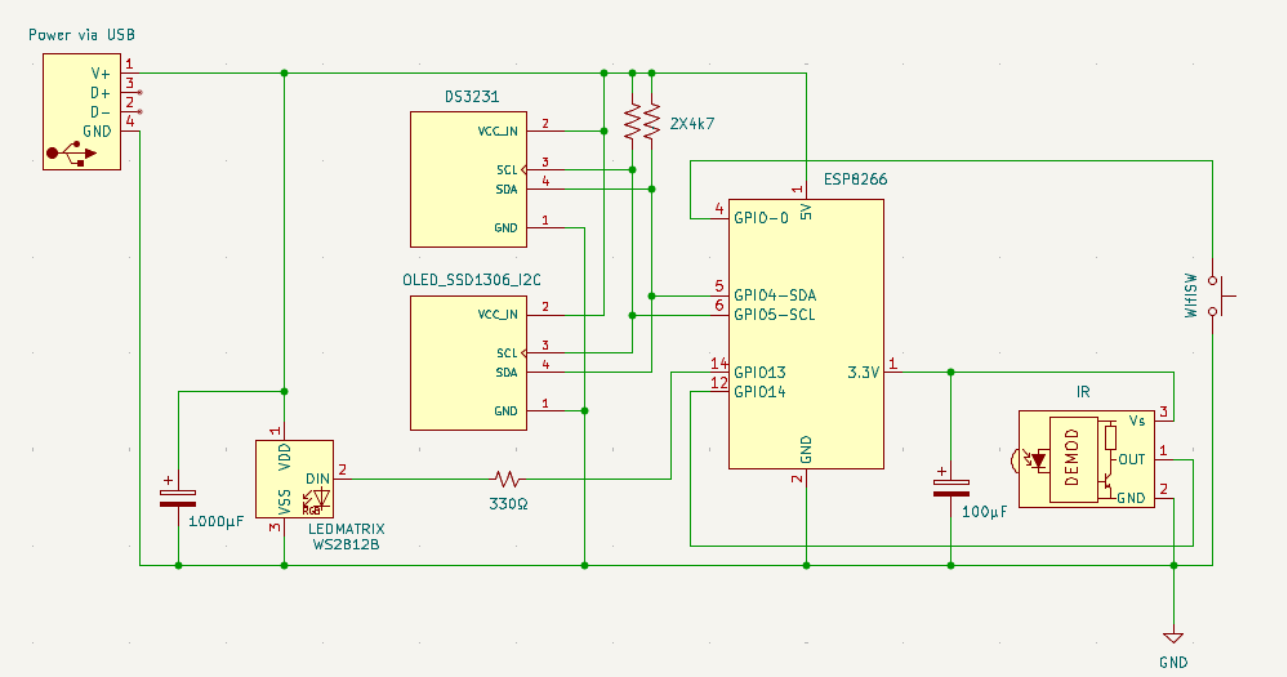
1. System overview

The final firmware is an ESP8266 clock controller integrating a 256-pixel WS2812 display, DS3231 RTC, SSD1306 OLED, direct NEC IR decoder, EEPROM configuration, bounded WiFi/NTP sessions, public-holiday caching, weekend rules and the new Animated daytime color mode.

2. Architecture and pins

Subsystem	**Role**
ESP8266	State machine, display, editor, networking and IR decode
DS3231	Continuous time source
WS2812 32 x 8	Primary display
SSD1306 128 x 64	Configuration/status UI
NEC IR	Primary user input
GPIO0 button	Boot-time WiFiManager setup
EEPROM 128 bytes	Persistent settings and holiday cache
Signal	**GPIO**
WS2812 data	13 / D7
IR receiver	14 / D5
Config button	0 / D3
OLED/RTC SDA	4 / D2
OLED/RTC SCL	5 / D1

3. Canonical schematic



4. Display and brightness

The logical matrix is 32 x 8. Standard digit positions are x=2, 8, 19 and 25 with DIGIT_Y=1. The central breathing separator uses columns 15 and 16. Brightness is 1-30 and fresh-install day brightness is 6. Night Mode remains the final brightness authority.

5. Timekeeping, WiFi and holidays

DS3231 is the continuous time source. Normal boot makes a bounded saved-network connection attempt. If successful, NTP updates the RTC, optional public-holiday data is refreshed, and WiFi is shut down. GPIO0 low at boot enters WiFiManager. AUTOSYNC default is 24 hours.

Country and timezone remain intentionally separate. Holiday rules independently support nationwide Public Holidays, Saturdays and Sundays; a matching rule changes only the effective Night Mode end time.

6. Seasonal Siesta engine

Season	Day	Night	Calendar
Summer	15:00-17:30	23:00-07:00	01/04-30/09
Winter	15:30-17:30	22:00-07:30	01/10-31/03

7. Color mode engine

Value	Mode	Behavior
0	MONO	Selected fixed user color
1	SOLID	One daily generated color
2	COLORFUL	Separate daily digit/accent colors
3	FUNNY	Fixed per-pixel palette

		permutation
4	DYNAMIC	Existing time-dependent dynamic color behavior
5	ANIMATED	Per-pixel smooth interpolation toward independently changing pseudo-random palette targets

Animated is inserted in the OLED Multi Color list immediately after Funny and before Dynamic. Dynamic keeps its previous numeric value 4, preserving existing EEPROM selections.

Animated refresh is limited to approximately 25 fps (40 ms) only when the editor is inactive and Animated mode is selected. Each transition segment is 1800 ms and uses smoothstep interpolation. A pixel's next segment starts exactly from its previous target, producing continuous motion without a short repeating sequence.

Existing priority rules remain unchanged: editor purple override, Siesta daytime orange and Night Mode behavior take precedence over normal daytime color modes.

8. EEPROM map

Address	Content
0-3	Night Mode start/end
4	Day brightness
5	Legacy timezone byte
6-9	Reserved legacy Siesta
10	Mono color
11	AUTOSYNC hours
12-14	Unused
15-16	Day color mode + magic
17-41	Seasonal Siesta + magic
42-44	12/24, font, display magic
45-46	OLED language + magic
47	Country
48	Holidays master
49-50	Holiday override HH:MM
51	Holiday settings magic
52-55	Holiday cache metadata
56-101	366-day holiday bitmap
102-104	Extended timezone + magic
105	Holiday rule mask
106	Holiday rule magic
107-127	Unused

9. Direct NEC remote

The final firmware keeps the direct NEC decoder and requires no IRremote dependency. The separate IR_Remote_Tester.ino helper intentionally uses Arduino-IRremote only as a temporary diagnostic utility for alternative remotes.

10. Dependencies

- ESP8266WiFi.h
- ESP8266HTTPClient.h

- WiFiClientSecureBearSSL.h
- TZ.h
- WiFiManager.h
- NTPClient.h
- WiFiUdp.h
- Adafruit_NeoPixel.h
- EEPROM.h
- U8g2lib.h
- RTClib.h

11. Source identity

Property	Value
Filename	Siesta_WiFi_Clock_HOLIDAYS_WEEKENDS_ANIMATED_FINAL.ino
Lines	8,210
Bytes	182,412
SHA-256	8f3d867dc6df555c29daf55e5889b521067dd3d46866f3558864752d70b486a9